Magdo Lillas

Giles

(a) $\frac{dM_{sys}}{dt} = In - Out$

$In = \dot{M}_i$

$Out = \dot{M}_e$

$\frac{dM_{sys}}{dt} = \dot{M}_i - \dot{M}_e$

$\frac{dM_{sys}}{dt} = 10 - 15$

$\int dM_{sys} = \int -5\, dt$

$M_{sys} = -5t + C$ 3/

$M(0) = 100\, kg$

$100\, kg = C$

$\boxed{M(t) = -5t + 100}$

$M(10) = -5(10) + 100$

$= -50 + 100$

$\boxed{M(10) = 50\, kg}$

great!

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(c)

$\frac{dM_A}{dt} = \dot{M}_{i,A} - \dot{M}_{e,A}$

$M_A = MW_A$

$= 10 \frac{kg\, tot}{min}\left(0.1 \frac{kg\, A}{kg\, tot}\right) - 15 \frac{kg\, tot}{min}\left(\frac{w(t)}{(100-5t)} \frac{kg\, A}{kg\, tot}\right)$

why?

$\frac{dM_A}{dt} = 1 \frac{kg\, A}{min} - \frac{15\, w(t)}{(100-5t)} \frac{kg\, A}{min} \Rightarrow \frac{dM_A}{dt} = 1 - \frac{3\, w(t)}{20-t}$

$\int dM_A = \int 1 - \frac{3}{20-t} w(t)\, dt$

$M_A(t) = t - 3\ln|20-t|\, w(t) + C \Rightarrow 60\, kg A = 0 - 3\ln|20|\, w(0)$

$w(0) = \frac{60}{3\ln|20|} \Rightarrow 6.68\, kg A$

$w(0) =$

$w(0) = 60 \frac{kg\, A}{kg\, tot}$

$M_A(0) = 60\, kg\, A$

Use $M_A = w M_{sys}$